

Chapter 12: Ratios, rates and time

Mixed revision of all 11 lessons · 28 questions

NSW syllabus outcome

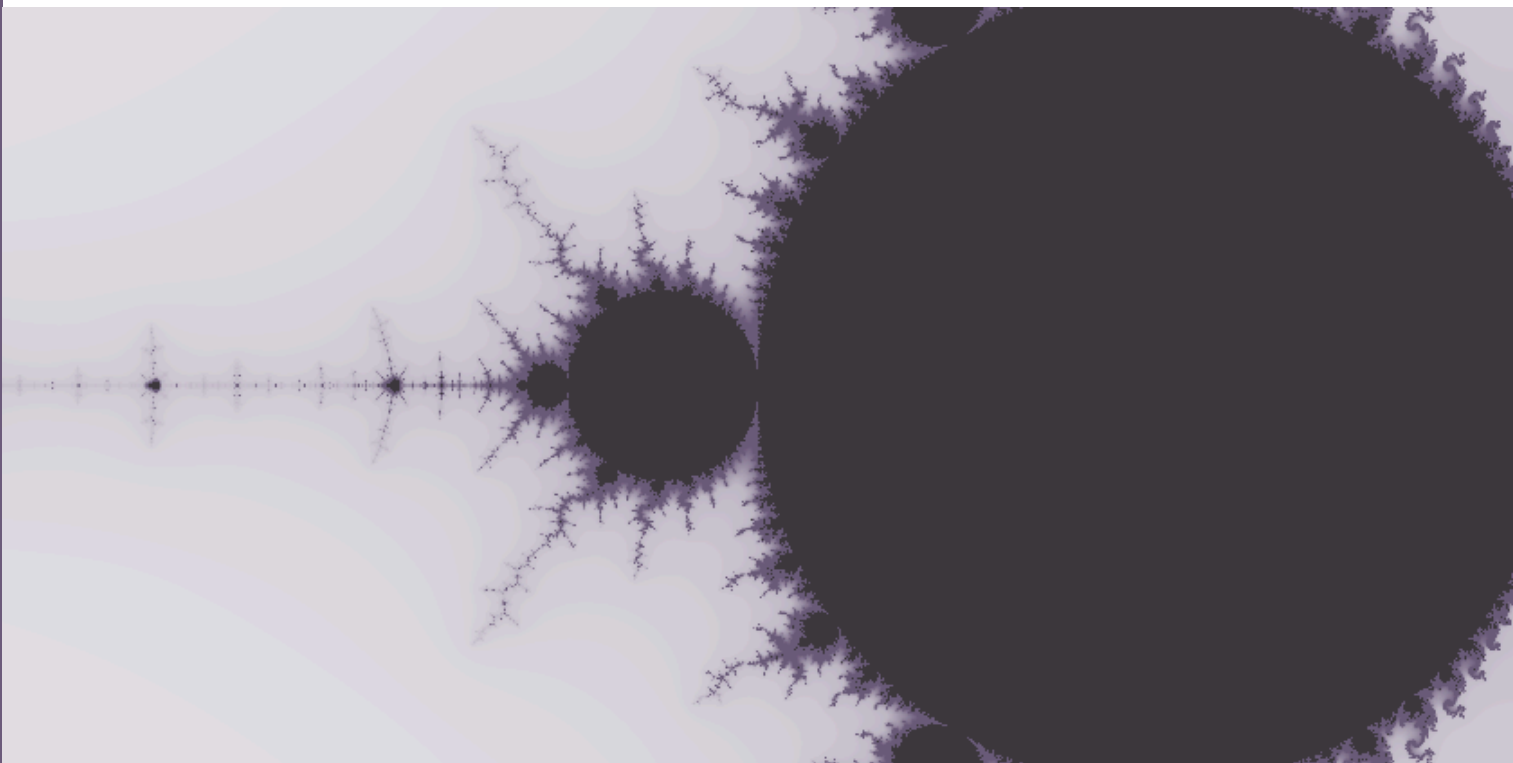
Stage 4 Number and algebra: MA4-RAT-C-01 solves problems involving ratios and rates, and analyses distance–time graphs; time calculations from Stage 3 extended.

NAME

CLASS

CHAPTER

12



Mixed revision homework. These questions mix all the lessons in Chapter 12, from **Easy** to **Challenging**. Each one is tagged with its lesson, so you can look back at that lesson's summary if you get stuck. Show your working in the boxes.

YOU DO

On your own: mixed revision.

QUESTIONS 1–10 OF 28

1 **Easy**

A bag has 8 red and 5 blue marbles. Write the ratio.
all to red Lesson 12.01

2 **Easy**

Convert.
6:25 p.m. to 24-hour Lesson 12.09

3 **Easy**

Find how many or how long.
1000 L at 40 L/min Lesson 12.06

4 **Easy**

Share \$60 in the ratio.
1 : 4 Lesson 12.03

5 **Easy**

How long is the trip?
10:35 → 12:10 Lesson 12.11

6 **Easy**

Convert.
200 min to h and min Lesson 12.08

7 **Easy**

Find the unit price.
8 for \$10 Lesson 12.05

8 **Easy**

Simplify the ratio.
16 : 24 Lesson 12.02

9 **Easy**

Find the time difference.
10:40 a.m. to 3:15 p.m. Lesson 12.10

10 **Easy**

Find the speed.
90 km in 2 h Lesson 12.07

YOU DO On your own: mixed revision.

QUESTIONS 11–20 OF 28

11 **Easy**

Write as a rate per 1 unit.

500 m in 100 s Lesson 12.04

12 **Medium**

Write each ratio.

A 9 m pole and a 4 m pole: short to long Lesson 12.01

13

Medium Convert 8:05 p.m. to 24-hour time.

Lesson 12.09

14

Medium How long to travel 180 km at 60 km/h?

Lesson 12.06

15

Medium Share \$80 in the ratio 3 : 5. Lesson 12.03

16

Medium

Use the bus timetable.

Write the times for the 16:10 bus in 12-hour time.

Lesson 12.11

Stop	Bus times (24-hour)		
School	15:10	15:40	16:10
Library	15:18	15:48	16:18
Shops	15:31	16:01	16:31
Beach	15:45	16:15	16:45

17

Medium Evaluate 2 h 40 min + 1 h 35 min.

Lesson 12.08

18

Medium Which is better: 1 L for \$3 or 2 L for \$5.40?

Lesson 12.05

19

Medium Simplify 16 : 40. Lesson 12.02

20

Medium Find the time from 22:30 to 01:10.

Lesson 12.10

YOU DO On your own: mixed revision.

QUESTIONS 21–28 OF 28

- 21** **Medium** A graph goes from 0 km at 1 p.m. to 30 km at 3 p.m. Find the speed. Lesson 12.07

- 23** **Challenging** A juice is 1 part concentrate to 6 parts water. Write the ratio of water to the whole drink. Lesson 12.01

- 25** **Challenging** Which is better: 250 g for \$2.50 or 1 kg for \$9? Lesson 12.05

- 27** **Challenging** Convert 12:45 a.m. to 24-hour time. Lesson 12.09

- 22** **Medium** A car travels 150 km in 2 h. Find its speed. Lesson 12.04

- 24** **Challenging** Juice : water = 1 : 4. How much water for 250 mL of juice? Lesson 12.03

- 26** **Challenging**
Think it through.
What does a line going down to the time axis show? Lesson 12.07

- 28** **Challenging**
Make your own timetable.
A ferry leaves the wharf every 40 minutes from 06:50 until 09:30. The trip takes 25 minutes. Write a timetable showing departure and arrival times. Lesson 12.11

Departs	Arrives